SDUHSD CAREER TECHNICAL EDUCATION

Digital Electronics

Campus: Canyon Crest Academy
CTE Industry Sector: Engineering & Architecture
CTE Pathway: Engineering Technology

Course Sequence: Course 3 - Capstone
Prerequisites: Intro to Engineering; recommended AP Computer Science A or AP Computer Science Principles
10 credits, UC Approved Area G
Recommended Pathway Sequence:

Course 1 - Introductory	Course 2 - Concentrator	Course 3- Capstone	Enrichment
Introduction to Engineering Techniques	Digital Electronics	Advanced Engineering & Technology	Robotics & Engineering Clubs

Brief Course Description:

- Standards for Engineering and Architecture
- http://www.corestandards.org/ELA-Literacy/RST/11-12/
- http://www.nextgenscience.org

Course Syllabi:

- **CCA**

Grading: A variety of assessments will be used including journal entries, projects and self and peer assessments. For example students will be assessed on their journal entries which include warm-up exercises to the 3dB point on a filter. Another example assessment includes student demonstration of a project implementing the combinatorial logic associated with a fire alarm. Another example assessment includes a student self-evaluation of the design of his or her sequential timer. Another example assessment includes peer review and voting of students' final class project.

Work load: (High, Medium, Low)

Class setting: This relates directly to how much the student is expected to work independently or in a group requiring self-initiated work, problem solving, etc versus and highly guided and structured class).
(Highly independent, Moderately Independent, Structured). Hands-on projects accompany each unit.
*Each year all CTE courses are reviewed by industry professionals within the pathway to ensure the most current, relevant experience for our students

Textbooks or Supplemental Materials for this course: Supplemental Materials provided electronically.

Software, program knowledge specific to this course: Scilab, National Instrument Multisim, Saleae Logic Analyzer

Industry Connections: Every February - students showcase projects with local businesses and community members

Units of Study, Large Projects, Work-based Learning Connections.

1. Fundamentals of Analog and Digital Electronics
2. Combinational Logic (Fire Alarm Control System Project)
3. Sequential Logic (Timer project)
4. FPGA Design (Garage Door Opener Control System Project)
5. Advanced Topics (Digital Filter, PN generator, Advanced Control Systems)

The following Model Curriculum Standards connected to this course:

Standards for Career Ready Practice
1. Apply appropriate technical skills and academic knowledge. 4. Apply technology to enhance productivity. 5. Utilize critical thinking to make sense of problems and persevere in solving them. 10. Demonstrate creativity and innovation.

Framework Concept	Engineering Technology
Fundamentals of Analog and Digital Electronics	● B1.0: Communicate and interpret information clearly in industry-standard visual and written formats. ● B3.0 Identify the fundamentals of the theory, measurement, control, and applications of electrical energy, including alternating and direct currents.
Combinational Logic	● B2.0 Demonstrate the sketching process used in concept development. ● B3.0 Identify the fundamentals of the theory, measurement, control, and applications of electrical energy, including alternating and direct currents. ● B4.0 Understand the concepts of physics that are fundamental to engineering technology.
Sequential Logic	● B7.0 Understand industrial engineering processes, including the use of tools and equipment, methods of measurement, and quality assurance. ● B8.0 Understand fundamental control system design and develop systems that complete pre-programmed tasks.
FPGA Design	● B7.0 Understand industrial engineering processes, including the use of tools and equipment, methods of measurement, and quality assurance. ● B8.0 Understand fundamental control system design and develop systems that complete pre-programmed tasks.
Advanced Topics	● B8.0 Understand fundamental control system design and develop systems that complete pre-programmed tasks. ● B9.0 Understand the fundamentals of systems and market influences on products as they are developed and released to production.